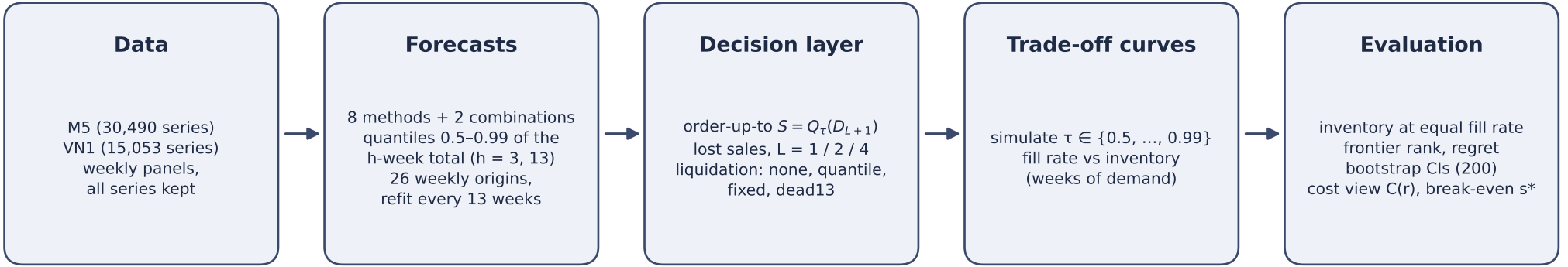
Data
M5 (30,490 series)
VN1 (15,053 series)
weekly panels,
all series kept
Forecasts
8 methods + 2 combinations
quantiles 0.5–0.99 of the
h-week total (h = 3, 13)
26 weekly origins,
refit every 13 weeks
Decision layer
order-up-to $S = Q_\tau(D_{L+1})$
lost sales, L = 1 / 2 / 4
liquidation: none, quantile,
fixed, dead13
Trade-off curves
simulate τ ∈ {0.5, …, 0.99}
fill rate vs inventory
(weeks of demand)
Evaluation
inventory at equal fill rate
frontier rank, regret
bootstrap CIs (200)
cost view C(r), break-even s*
Repeated on 3 test windows (offset 0 / 26 / 52 weeks), 3 lead times, 4 hyper-parameter settings